Analysis of North Carolina's 2013 Unemployment Insurance Reduction

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Review the Research Question & Project Positioning

- This paper aims to understand the effect that Unemployment Insurance has on the Unemployment rate.
- The paper analyzes the effect of North Carolina Unemployment Insurance (UI) Policy Change.
- The policy cuts UI down from 26 weeks to a maximum of 20 weeks and a minimum of 12 weeks based on Unemployment Rate.
- This policy was a controversial purely political decision by the North Carolina Governor.

Presentation of the Research Methodology

- Found the unemployment rate of North Carolina.
- Found a state with a similar unemployment rate before the policy change with similar demographics, South Carolina.
- Compared the unemployment rate of North Carolina and South Carolina from 2001-2020.
- Assessed the unemployment rate before and after the policy change.

Presentation of Raw Data

Period	State	Mean	Median	Max	Min
2001-2008 (Pre-Recession)	NC	5.89	5.70	9.6	4.4
	SC	6.73	6.90	9.4	4.1
2008-2013 (Pre-Policy)	NC	9.24	9.80	11.4	5.2
	SC	9.48	9.65	12.1	6.0
2013-2020 (Policy Change)	NC	5.54	5.15	14.2	3.8
	SC	5.10	5.05	11.8	2.4

Difference in Difference Regression

$$\text{Unemployment_Rate} = \beta_0 + \beta_1 \text{Treatment} + \beta_2 \text{Post} + \beta_3 (\text{Treatment} \times \text{Post}) + \epsilon$$

Difference in Differences

```
Call:
lm(formula = Unemployment_Rate ~ Treatment + Post + Treatment:Post,
    data = df_long_filtered)

Residuals:
    Min      1Q  Median      3Q     Max
-3.9056 -1.5882 -0.5882  1.3994  4.1143

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)      8.0056     0.1605   49.89  < 2e-16 ***
Treatment       -0.6174     0.2269   -2.72  0.00677 **
Post            -3.0258     0.2644  -11.45  < 2e-16 ***
Treatment:Post   0.9233     0.3739    2.47  0.01390 *
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.926 on 452 degrees of freedom
Multiple R-squared:  0.3031,  Adjusted R-squared:  0.2985
F-statistic: 65.53 on 3 and 452 DF,  p-value: < 2.2e-16
```
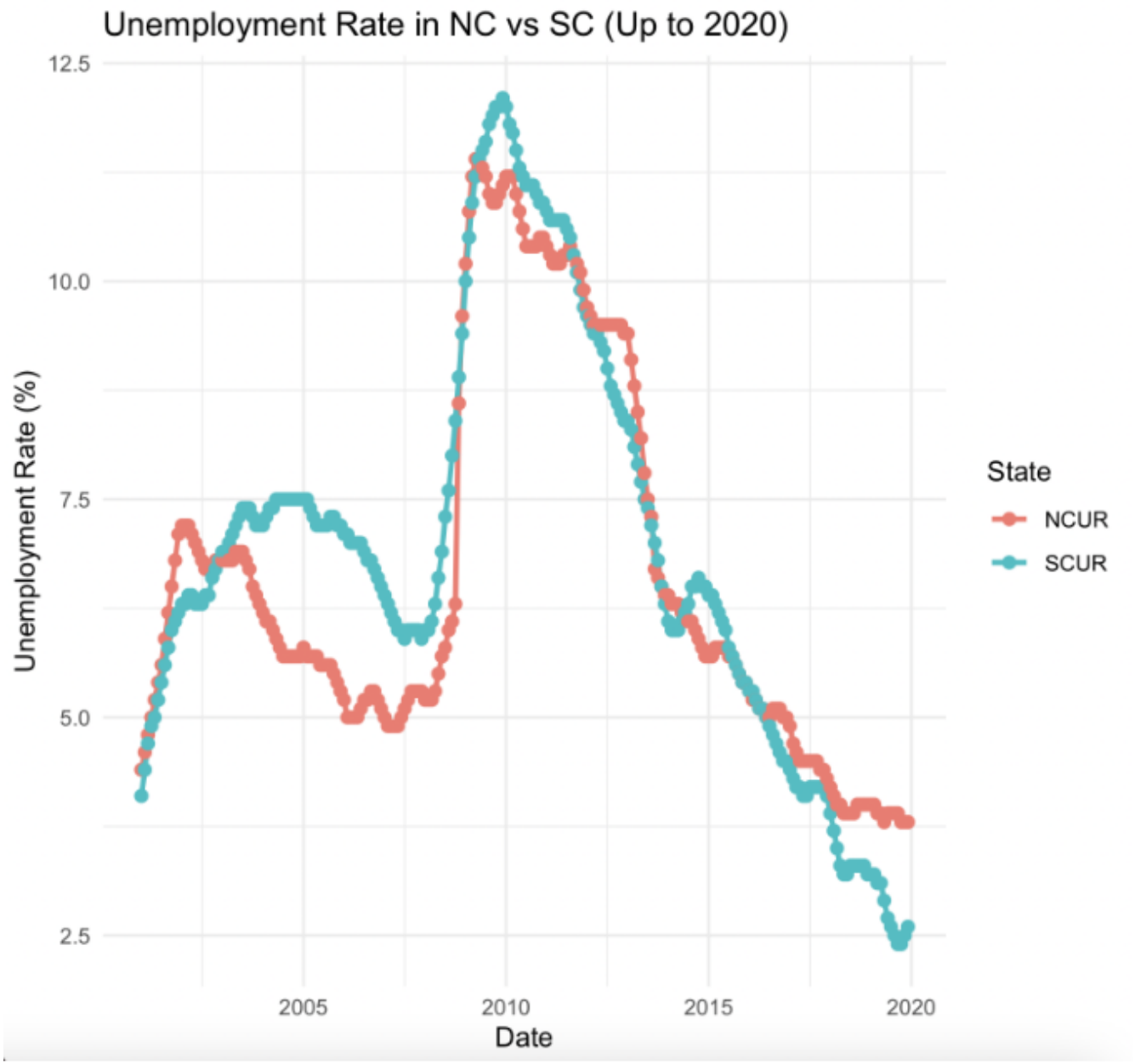

Presentation of Results: Time Series



Conclusion

- **Difference in Differences:**
  - There was no significant difference between North Carolina's unemployment rate compared to South Carolina from 2000-2020. $p = 0.1250$
  - However, there was a significant difference in unemployment before vs. after 2013. Many confounding variables exist, most notably that the US was recovering from the 2008 Recession. $p < 0.001$
  - The most significant result was the difference in how North Carolina's unemployment changed compared to South Carolina's. The p-value from the DiD regression was $p = 0.0139$, suggesting a meaningful difference in how North Carolina's unemployment rate changed compared to South Carolina's.

Conclusion Pt.2

- This regression suggests that North Carolina's unemployment rate recovered more slowly than South Carolina's after the policy change.
- More research is needed to identify confounding variables and macroeconomic factors to establish a causal relationship between the policy change and unemployment.
- Additionally, comparing NC to more states across the US would strengthen the analysis.
- However, these results suggest that the policy may have been counterproductive to the governor's goal of reducing unemployment.

Citations

- Evermore, Michele. “North Carolina’s Unemployment Insurance System: Unprepared for Recession, despite Solvent Trust Fund.” National Employment Law Project, 28 Apr. 2024. www.nelp.org/insights-research/north-carolinas-unemployment-insurance-system-unprepared-f
- “Unemployment Rate in North Carolina.” FRED, 5 Mar. 2025. fred.stlouisfed.org/series/NCUR